

CSCI 340, Winter 2023, Math Homework # 1

Trey Custodio

Show your work. Explain, in each case, how you got a particular numerical result. You may, of course, use a computer for numeric computations, but which expressions you used to get each result must be made clear. For example, you can't just put down 12, you must put down $2^2 + 2^2 + 2^2 = 12$. You can't just put down 4.443, you must put down $\pi \cdot \sqrt{2} = 4.443$

If you have trouble writing neat math, consider learning and using L^AT_EX.

1. Let $v = \langle 2, 2, 1 \rangle$ and $w = \langle 1, -2, 0 \rangle$. Find the following:

(a) $v \cdot w$

$$v \cdot w = \sum_{i=0}^n (v[i] + w[i])$$

$$\begin{aligned} \text{So } v \cdot w &= (2 * 1) + (2 * (-2)) + (1 * 0) \\ &= (2) + (-4) + (0) \\ &= -2 \end{aligned}$$

(b) The vector projection of w on v .

$$\begin{aligned} Proj &= \frac{v \cdot w}{|w|^2} * w, \quad |w|^2 = (1)^2 + (-2)^2 + 0^2 = 1 + 4 = 5 \\ Proj &= \frac{-2}{5} * w = \langle 1 * \frac{-2}{5}, (-2) * \frac{-2}{5}, 0 * \frac{-2}{5} \rangle \\ &= \langle \frac{-2}{5}, \frac{4}{5}, 0 \rangle \end{aligned}$$

2. Normalize each of the following vectors, in other words, make each vector unit length.

(a) $v_1 = \langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0 \rangle$

$$Unit_v = \frac{v}{|v|}$$

$$|v_1| = \sqrt{(\frac{\sqrt{2}}{2})^2 + (\frac{\sqrt{2}}{2})^2 + (0)^2} = 1$$

$$\begin{aligned} \text{So } U_{v_1} &= \frac{v_1}{1} = \langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, \frac{0}{1} \rangle \\ &= \langle \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0 \rangle = v_1 \end{aligned}$$

(b) $v_2 = \langle -1, 1, -1 \rangle$

$$|v_2| = \sqrt{(-1)^2 + (1)^2 + (-1)^2} = \sqrt{3}$$

$$\text{So } U_{v_2} = \langle \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}} \rangle$$

(c) $v_3 = \langle 0, -2, -2 \rangle$

$$|v_3| = \sqrt{(0)^2 + (-2)^2 + (-2)^2} = \sqrt{8}$$

$$\text{So } U_{v_3} = \langle \frac{0}{\sqrt{8}}, \frac{-2}{\sqrt{8}}, \frac{-2}{\sqrt{8}} \rangle = \langle 0, \frac{-2}{\sqrt{8}}, \frac{-2}{\sqrt{8}} \rangle$$

3. Find the cosine of the angle between the vectors $v = \langle 1, 2, 3 \rangle$ and $w = \langle 3, 2, 1 \rangle$

$$v \cdot w = \cos(\theta) |v| |w|$$

$$v \cdot w = (1 * 3) + (2 * 2) + (3 * 1) = (3) + (4) + (3) = 10$$

$$|v| = \sqrt{(1)^2 + (2)^2 + (3)^2} = \sqrt{1 + 4 + 9} = \sqrt{14}$$

$$|w| = \sqrt{(3)^2 + (2)^2 + (1)^2} = \sqrt{9 + 4 + 1} = \sqrt{14}$$

$$\cos(\theta) = \frac{v \cdot w}{|v| |w|}, \quad |v| |w| = \sqrt{14}^2 = 14, \quad v \cdot w = 10$$

$$\cos(\theta) = \frac{10}{14} = \frac{5}{7}$$

4. For the set of vectors and points on the next page, find:

- (a) The coordinates of q in the frame $\langle p, u, v \rangle$
 $q : (9, 3), \quad p : (3, 7), \quad u : (2, 0), \quad v : (-4, 4)$

$$(X): \quad q_x = p_x + au_x + bv_x, \quad 9 = 3 + 2a - 4b$$

$$(Y): \quad q_y = p_y + au_y + bv_y, \quad 3 = 7 + 0a + 4b$$

$$(b): \quad 3 = 7 + 4b, \quad -4 = 4b, \quad b = -1$$

$$(a): \quad 9 = 3 + 2a - 4(-1), \quad 6 = 2a + 4, \quad 2 = 2a \quad a = 1$$

$$q = (1, -1)$$

- (b) The coordinates of q in the frame $\langle p, u, w \rangle$
 $q : (9, 3), \quad p : (3, 7), \quad u : (2, 0), \quad w : (2, 2)$

$$(X): \quad 9 = 3 + 2a + 2b$$

$$(Y): \quad 3 = 7 + 0a + 2b$$

$$(b): \quad 3 = 7 + 2b, \quad -4 = 2b, \quad b = -2$$

$$(a): \quad 9 = 3 + 2a + 2(-2), \quad 6 = 2a - 4 \quad 10 = 2a, \quad a = 5$$

$$q = (5, -2)$$

- (c) The coordinates of q in the frame $\langle p, v, w \rangle$
 $q : (9, 3), \quad p : (3, 7), \quad v : (-4, 4), \quad w : (2, 2)$

$$(X): \quad 9 = 3 + (-4)a + 2b$$

$$(Y): \quad 3 = 7 + 4a + 2b$$

$$(X + Y): \quad 12 = 10 + 0a + 4b$$

$$(b): \quad 2 = 4b, \quad b = \frac{1}{2}$$

$$(a): \quad 3 = 7 + 4a + 2\left(\frac{1}{2}\right), \quad -4 = 4a + 1, \quad -5 = 4a, \quad a = -\frac{4}{5}$$

$$q = \left(-\frac{4}{5}, \frac{1}{2}\right)$$

- (d) The coordinates of p in the frame $\langle q, u, v \rangle$
 $p : (3, 7), \quad q : (9, 3), \quad u : (2, 0), \quad v : (-4, 4)$

$$(X): \quad 3 = 9 + 2a - 4b$$

$$(Y): \quad 7 = 3 + 0a + 4b$$

$$(b): \quad 4 = 4b, \quad b = 1$$

$$(a): \quad 3 = 9 + 2a - 4(1), \quad -6 = 2a - 4, \quad -2 = 2a, \quad a = -1$$

$$p = (-1, 1)$$

- (e) The coordinates of p in the frame $\langle q, u, w \rangle$
 $p : (3, 7), \quad q : (9, 3), \quad u : (2, 0), \quad w : (2, 2)$

$$(X): \quad 3 = 9 + 2a + 2b$$

$$(Y): \quad 7 = 3 + 0a + 2b$$

$$(b): \quad 4 = 2b, \quad b = 2$$

$$(a): \quad 3 = 9 + 2a + 2(2), \quad 3 = 2a + 13, \quad -10 = 2a, \quad a = -5$$

$$p = (-5, 2)$$

- (f) The coordinates of p in the frame $\langle q, v, w \rangle$
 $p : (3, 7), \quad q : (9, 3), \quad v : (-4, 4), \quad w : (2, 2)$

$$(X): \quad 3 = 9 - 4a + 2b$$

$$(Y): \quad 7 = 3 + 4a + 2b$$

$$(X + Y): \quad 10 = 12 + 0a + 4b$$

$$(b): \quad -2 = 4b, \quad b = -\frac{1}{2}$$

$$(a): \quad 3 = 9 - 4a + 2(-\frac{1}{2}), \quad 3 = 9 - 4a - 1, \quad -7 = -4a, \quad a = \frac{7}{4}$$

$$p = (\frac{7}{4}, -\frac{1}{2})$$

